9-4-26 Sujit class 1

Decision Theory: ch 12, 13

A brief intro

A pen is being sold at a shop for ₹100
What makes you decide to buy this pen?
Capture this for autonomous agent?
Techniques to take decisions in the face of uncertainty.

When faced with uncertainty (normal in any real world task), dont use ad-hoc ~~tas~~ techniques.

Decision theory provides well-developed techniques.

But, what is uncertainty, and why do we need to reason with it?

Which route to take?

Route 1: Sibiu → Fagaras → Bucharest : 310
Route 2: Rimnicu → Pitesi → Bucharest 278
Ideally would take shortest route
What if traffic is a factor?

Suppose you have accurate info that:
R1 has fast moving traffic while R2 has slow moving traffic.

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Date ______
Page ______

A logical agent may reason

Slow (R2) fast(R1)

Slow(x) ⇒ Avoid(x)

Avoid(x) ^ fast(y) ⇒ select(y)

Agent selects R1

Do you think this is a good way to reason?
(Why/ Why not?

		On time		Delay (x% of time)		Time
R1	SFB	80%	4h	20%	10h	5.2h
R_2	SRPB	70%	4.5h	30%	5h	<5h
						80%

How much do you care about uncertainty

Uncertainty:

May not have categorical answers

May not know exactly which route is slow.

50% chance that one route is slow

Even if we know slow, how slow!

Changes the way agent takes decision

Probability theory used in helping agents reason with uncertainty

- Summarize uncertainty due to our ignorance or laziness
- may be very costly to remove such uncertainty

~~Decision Theory~~
~~(How to make~~

Rational decisions depend on
Relative importance of various goals
Liklihood or degree to which they will be achieved

Decision Theory
(How to make decisions)

Decision Theory = probability Theory + utility Theory
(Deals with chance) (Deals with outcomes)

Fundamental idea:
The MEU (Maximum expected utility) principle
Agent is rational iff it chooses the action that yields highest expected utility averaged over possible outcomes of the action

Weigh the utility of each outcome by the probability that it occurs.

Utility Theory: Von Neumann (Game Theory)

Let X be the set of outcomes. $\succ$ be the preference of a player over the set of outcomes

Axioms:

- Completeness: every pair of outcomes is ranked
- Transitivity: If $x_1 \succ x_2$ & $x_2 \succ x_3$ then $x_1 \succ x_3$
- Substitutability: If $x_1 \sim x_2$ then any lottery in which x_1 is substituted by x_2 is equally preferred
- Decomposibility: two different lotteries assign same probability to each outcome, then player is indifferent between these two lotteries
- Monotonicity: If $x_1 \succ x_2$ & $p > q$ then $[x_1 : p, x_2 : 1-p] \succ [x_1 : q, x_2 : 1-q]$
- Continuity: If $x_1 \succ x_2 \succ x_3$, $\exists p \ni x_2 \sim [x_1 : p, x_3 : 1-p]$

Von Neumann & Morgenstern

Theorem:
Given a set of outcomes X & a preference relation on X that satisfies above six axioms, there exists a utility function $u: X \rightarrow [0,1]$ with the following properties:

1) $u(x_1) \geq u(x_2)$ iff $x_1 > x_2$

2) $U([x_1 : p_1 ; x_2 : p_2 ; \dots ; x_m : p_m])$

$= \sum_{j=1}^{m} p_j \, u(x_j) \qquad \sum p_j = 1$

Basic Probability Notation

Probabilities over sample space of a variable - $P(X)$

Called the probability of distribution for random variable

Map of sample space to real numbers

$P(\text{Weather} = \text{sunny}) = 0.6$
$P(\text{Weather} = \text{rain}) = 0.1$
$P(\text{Weather} = \text{cloudy}) = 0.29$
$P(\text{weather} = \text{snow}) = 0.01$
$P(\text{weather}) = \langle 0.6, 0.1, 0.29, 0.01 \rangle$

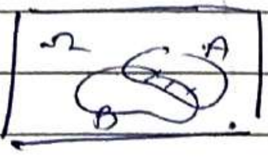
$\sum P(X = x_i) = 1$
$x_i \in \{\text{sunny}, \text{rain}, \text{cloudy}, \text{snow}\}$

Conditional Probability (Posterior)

$A, B \qquad P(A|B) = \dfrac{P(A \wedge B)}{P(B)} \qquad$ if $P(B) > 0$

n
$k_1 \rightarrow B$
$k \rightarrow A \cap B$

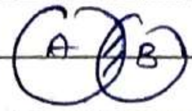
$$k_1 / k = \frac{k_1/n}{k/n} = \frac{P(A \wedge B)}{P(B)}$$



$P(X = \text{sunny} \mid \text{last 2 days were sunny})$

If we know B & C, $P(A \mid B \wedge C)$

Product Rule:



$P(A \mid B) = P(A \wedge B) / P(B)$

$P(A \wedge B) = P(A \mid B)\, P(B)$

$P(A \wedge B) = P(B \mid A)\, P(A)$

Tootache → Cavity

Cavity → toothache?

Joint Probability Distribution

Notation for distributions on multiple variables

$P(\text{Weather}, \text{Cavity})$ i.e $P(\text{Weather} \wedge \text{Cavity})$

Joint PD is a table:

Assings prob to all possible assignment of values for combinations of variables

Weather has values (Sunny, R, C, Snow)

Cavity has values (T, F)

$4 * 2$ table of prob.

Inference using full joint distribution

3 var: {Cavity, toothache, Catch}

$2 * 2 * 2$ table

Dentist catches or not

$P(X_1, X_2, \ldots X_n)$ probs to all possible assignment of values to variables $X_1, X_2 \ldots X_n$

	toothache		¬toothache	
	catch	¬catch	catch	¬catch
cavity	.108	.012	.072	.008
¬cav	.016	.064	.144	.576

$P(\text{Cavity}) = 0.2 = 0.108 + .012 + .072 + .008$

Marginal prob of cavity

$$P(X=x) = \sum_{(y,z)} P(X=x, Y=y, Z=z)$$

Marginalization

$P(\text{cavity} \mid \text{toothache}) = \cancel{.108 + .012 = 0.12}$

$$\frac{P(C \wedge T)}{P(T)} = \frac{0.12}{0.2} = 0.6$$

$P(\neg\text{cavity} \mid \text{toothache}) = .4$

$= P(\neg\text{cav} \wedge \text{toothache}) / P(\text{ache})$

$P(c \mid \text{ache}) + P(\neg\text{cav} \mid \text{ache}) = 1$

Denominator can be viewed as normalization constant $1/P(\text{toothache})$

$P(\text{cav} \mid \text{ache}) = \alpha\, P(\text{cav}, \text{ache})$

$= \alpha\, [P(\text{cav}, \text{ache}, \text{catch}) + P(\text{cav}, \text{ache}, \neg\text{catch})] = .108 + .012$

$= \alpha\, 0.12$

α P(¬cav, toothache) = α [.016 + .064]
= α 0.08

Normalizing α <.12, .08> gives <.6, .4>

Issue: For domain with n Boolean var,
input table size $O(2^n)$

Full joint dis in tabular form not a practical tool for building reasoning systems.

Independence:
Helps in reducing size of domain representation & complexity of inference problem.

Lets add weather into example: sun, r, c,
snowy - 2 x 2 x 2 x 4 entries in table

P(ache, catch, cav, cloudy) = P(cloudy | toothache, catch, cav) P(ache, catch, cav)

P(cloudy | ache, catch, cav) = P(cloudy)

32 elements table becomes 8 elements +
4 elements table

$P(A \wedge B) = P(A)P(B)$
$P(A|B) = P(A)$

Baye's Rule:

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Given that

$P(A \cap B) = P(A|B)\, P(B)$

$P(A \cap B) = P(B|A)\, P(A)$

$$P(B|A) = \frac{P(A|B)\, P(B)}{P(A)}$$

P(effect | cause) P(A|B) causal knowledge

P(cause | effect) P(B|A) diagnostic knowledge

Often have good prob on casual relationships
→ P(symptoms | disease)
Need to infer diagnostic knowledge
many times.

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12-4-25

Baye's Rule.

$$\underset{\substack{\text{Posterior} \\ \text{Diagnostic}}}{P(B|A)} = \frac{\overbrace{P(A|B)}^{\text{Likelihood / Causal}}\, \underbrace{P(B)}_{\text{Prior}}}{\underset{\text{Evidence}}{P(A)}}$$

Bayes' Rule Example:

S: Proposition that patient has
stiff neck

M: Prop that he has menangitis

M- causes S 70% of time

Given
$P(S|M) = 0.7$
$P(M) = 1/50,000$
$P(S) = 0.01$

$P(M|S) = 0.0014 = P(S|M)\,P(M)/P(S)$

Why so low?

Bayes Rule use:

Why not estimate P(M|S) the same way we do P(S|M)?
why do we need Bayes Rule?

P(S|M) is causal knowledge does not change
It is "model based"
It reflects the way menangitis works

P(M|S) is diagnostic, tell us likelihood of M given symptoms
Diagnostic knowledge may change with circumstance, thus helpful to derive it
If there is an epidemic, prob of menin goes up, rather than trying to estimate P(M|S) which is very hard, we can compute it using other terms
However, P(S|M) is unaffected by the epidemic

Use of causal or model based knowledge provides crucial robustness for probabilistic systems.

Computing the Denominator P(S)

We wish to avoid computing the denominator in Bayes rule
May be hard to obtain
In our example, denom is P(S)
Techniques to compute (avoid computing P(S))

Our first approach is to compute relative likelihoods
If M (menan) & W (Whiplash) are 2 possible explanations:

$P(M|S) = P(S|M)\,P(M) / P(S)$

$P(W|S) = P(S|W)\,P(W) / P(S)$

$P(M|S) / P(W|S) = P(S|M)\,P(M) / P(S|W)\,P(W)$

Computing the denominator P(S)
Another approach Approach 2

$P(M|S) = P(S|M)\,P(M) / P(S)$

$P(NOT(M)|S) = P(S|NOT(M))\,P(NOT(M)) / P(S)$

$P(M|S) + P(NOT(M)|S) = 1$
(They must sum to 1)

$[P(S|M)\,P(M)/P(S)] + [P(S|NOT(M)) * P(NOT(M))/P(S)] = 1$

$P(S|M) * P(M) + P(S|NOT(M)) * P(NOT(M)) = P(S)$

calculate P(S) in this way

Simple Example
Suppose 2 identical urns
URN1 colored red from inside has 1/3 black 2/3 red balls
URN2 colored black from inside 1/3 red 2/3 black balls
we select 1 urn at random, cant tell how its colored inside
What is the probability that urn is colored red inside 0.5
What if we were to select ball at random from the urn & its red? Does that change the probability?

Apply Bayes Rule
$P(\text{Red urn} | \text{Red ball}) = P(RB | RU) * P(RU) / P(RB) = \frac{2}{3} \times 0.5 / P(\text{red ball})$

How to calculate P(red ball)?

$P(BU|RB) = P(RB | BU) * P(BU) / P(RB)$
$= 1/3 * 0.5 / P(RB)$

~~P(RB) = 0~~

By approach #2 $\frac{2}{3}(0.5) / P(RB)$

$+ \frac{1}{3} 0.5 / P(RB) = 1$

$P(RB) = 0.5$

$\therefore$ ~~P(RB)~~ $P(RU|RB) = 2/3$

More general forms of Bayes Rule

$P(Y|X) = P(X|Y) * P(Y)/P(X)$

Bayes rule for multi-valued variables
Generalize to some background evidence e

$P(Y|X,e) = P(X|Y,e) * P(Y|e) / P(X|e)$

Conditional Independence

toothache & catch are independent given presence of cavity

Each is directly caused by cavity, toothache is caused by cavity & dentist catches cavity using probe

A toothache can't be caught by a probe nor a probe results in a toothache

Conditional Independence captured as

$P(TA \wedge catch \mid cav) = P(TA \mid catch, cav)$
$P(catch \mid cav)$.

$P(TA \mid cat, cav) = P(A \mid B, C)$
If A & C are conditionally ind given B ~~then~~ prob of A is not dep on C
$P(A \mid B \wedge C) = P(A \mid B)$

$\therefore P(TA, catch \mid cav) = P(TA \mid cav)$
$P(catch \mid cav)$

Combining Evidence

Revisiting Menengitis example

S: Prop that patient has stiff neck
H: Prop that pat has severe headache
M: Prop " " " menen
m causes S, 50% of time
m causes H, 70% of time

Prob of menen should go up if both symptoms reported - how to combine such symptoms?

$P(A \mid B \wedge C) = P(A \wedge B \wedge C) / P(B \wedge C)$

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Date
Page

Numerators:

$P(M \wedge S \wedge H) = P(S \mid M \wedge H)\, P(M \wedge H)$
$= P(S \mid M)\, P(M \wedge H)$
Why?

$= P(S \mid M)\, P(H \mid M)\, P(M)$

Going back to our example

$P(M \mid S \wedge H) = P(S \mid M)\, P(H \mid M) * P(M) / P(S \wedge H)$

S & H are cond ind

Decision Theory

Revisiting Romania example

If P1 & P2 are two plans:
P1 uses R1 :
$P(\text{home-early} \mid P1) = 0.8$
$P(\text{stuck1} \mid P1) = 0.2$
R1 is quick if flowing but stuck for 1 hour if slow
$U(\text{home-early}) = 100$
$U(\text{stuck1}) = -1000$
Assigned numerical values to outcomes

P2 uses R2
$P(\text{home somewhat early} \mid P2) = 0.7$

classmate
Date
Page

P (stuck 2 | P2) = 0.3
R2 will be somewhat quick if flowing but not bad even if slow
U (home somewhat early) = 50
U (stuck2) = -10

Application of MEU example

EU (P1) = P(home early | P1) P U(home early)
+ P (stuck1 | P1) U(stuck1)

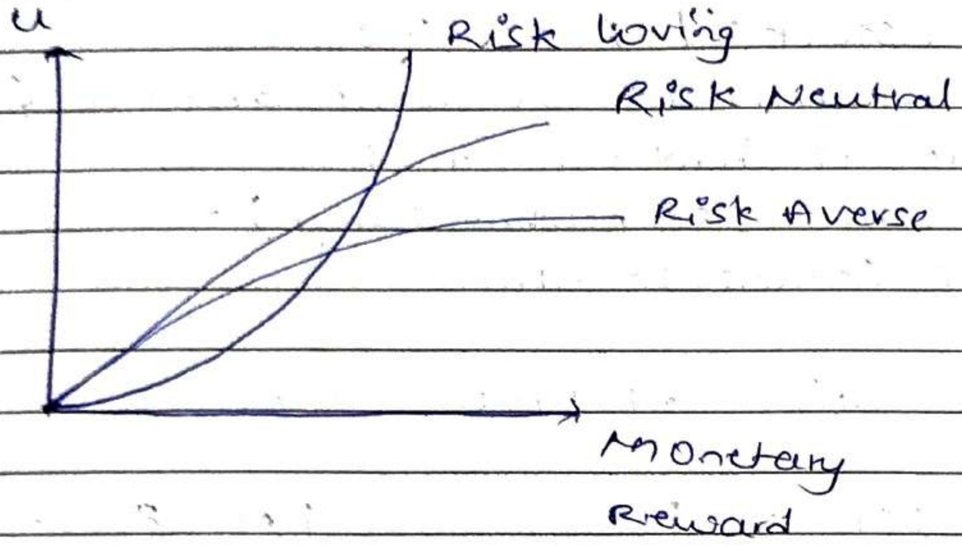
= 0.8 * 100 + 0.2 (-1000) = -120

EU(P2) = P(home somewhat early | P2)
U ~~P~~ (home somewhat early)
+ P (Stuck2 | P2) U(stuck2)

= 0.7(50) + 0.3(-10) = 32

Risk aversion
We are risk averse

Our utility functions for money are as follows:
First million means a lot (US $1M) = 10
Sec mil not so much = 15 (NOT 20)
Third mil even less so = 18 (NOT 30)



Additional money not buying as much utility
If we plot amt of money on x & utility on y we get concave curve

Risk Averse, Risk Neutral, risk seeking

More Risk Aversion
Key: Slope of utility function is continuously decreasing
we will refuse to play a monetarily fair bet

Suppose we start with x dollars
we are offered a game
0.5 chance to win 1000 dollar
(c = 1000)
0.5 to lose 1k (c = 1k)

Expected monetary gain/loss is 0

(hence monetarily fair)

Should be neutral to it, but seems we are not. Why?

$U(x+c) - U(x) < U(x) - U(x-c)$

$U(x+c) + U(x-c) < 2U(x)$

$\{U(x+c) + U(x-c)\}/2 < U(x)$

EU (playing game) < EU (not playing game)

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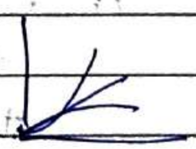
Recap Bayes Theorem

Conditional Independence



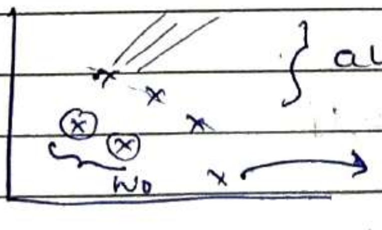
Utility Theory:

- Risk averse — concave
- Risk Neutral — straight
- Risk seeking / loving — convex

MEU.

Pareto Efficient

If their is no other better outcome



} all 3 are PE

will also become PE if its below the 4.

Weighted

ω_1 H_1 x_1 $\sum \omega_i EU_i$
ω_2 H_2 x_2 Pareto Efficient
$\vdots$ $\vdots$
ω_n x_n

Option x is pareto eff if ∃ no out come y for which on all ~~oth~~ aspects, y is better.

Pareto efficient:
Suppose we are dealing with N- dimensional utilities. Option x is pareto efficient if
$\nexists y$ st $U_i(y) \geq U_i(x)$ $i = 1, \cdots N$
with strict inequality for atleast one i

A	Production	Marketing
a_0	+	+
~~a_1~~	+	−
a_2	−	+
a_3	−	− → No change

A = action space

State: Modeling inventory
$S = \{L, M, H\}$

L M H

Transition Probability
$P: S \times S \times A \rightarrow [0,1]$
current × future × action

Reward $R: S \times S \times A \rightarrow \mathbb{R}$
current × destination / future state × action.

$P_a(s, s')$: Probability of system moving from state s to s' if we take action a

$R_a(s, s')$: The reward that we obtain if we take an action a in state s & reach state s'

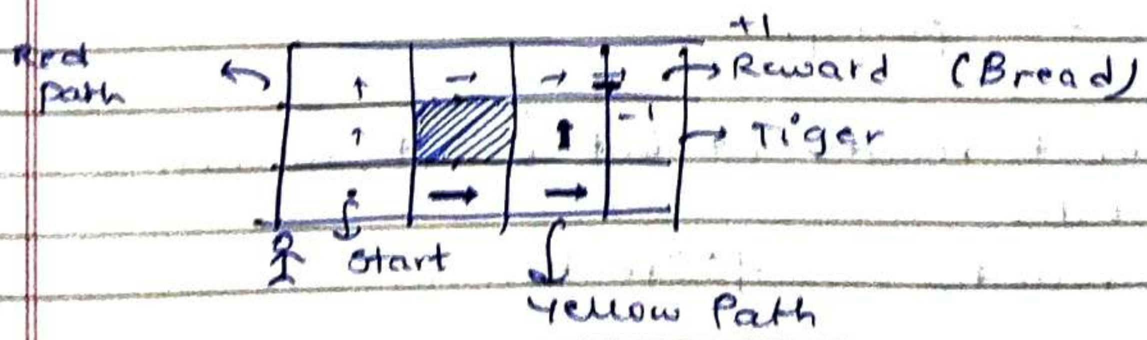
<S, A, R, P> tuple is called MDP
Markov Decision Process.

Markov Decision Process :

Based on current state only, not previous state.

Example: What sequence of actions can agent take?
Agent can take action N, E, S, W.
Each action costs : $-1/25$

Possible states
(1,3) (1,2) (1,1)
(2,3) (2,1)
(3,3) (3,2) (3,1)
(4,1) (4,2) (4,3)



Have to consider all future paths in utility
Simple search algo? Not deterministic

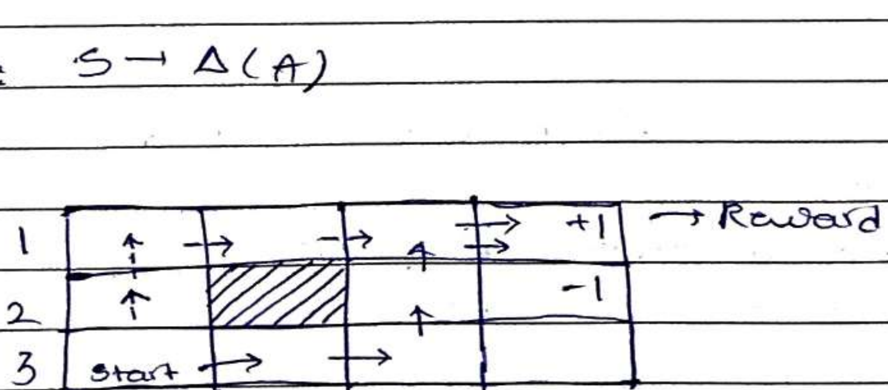
Red: N, N, E, E, E
Yellow: E E N N E

Have to consider state transitions. In each state decide & compare action.
Deterministic Policy:

$\pi : S \rightarrow A$

Randomized Policy to take actions:

$\pi : S \rightarrow \Delta(A)$



For MDP we can argue that no randomized policy performs better than optimum deterministic policy.

After t rounds we don't revert, no more states
T rounds finite horizon

If you interact infinitely many times → infinite horizon

At $T=0$ i use π^0 policy
$T=1$ π^1
$T=2$ π^2

If I use the same policy for all ~~ar~~ rounds - stationary policy

$$\pi^0 = \pi^1 = \pi^2 = \cdots$$

$|A|^{|S|}$ = # of policies possible for each round (each policy for all states)

~~Alor~~ For MDP, deterministic & stationary policy works.

Say I have 2 states ⓪ ① actions a_0, a_1

eg:

$\pi_0 - (0, a_0)\ (1, a_1)$

$\pi_1 = (0, a_1)\ (1, a_1)$

$\pi_2 = (0, a_1)\ (1, a_0)$

$\pi_3 = (0, a_0)\ (1, a_0)$

4 policies.

Focus on det., sta.

Finite horizon:

No more rewards ~~or~~ game over after T rounds.

T − 1 | T
①, ②, ⓢ, …, |S| | ① ($a_0, \ldots, a_{|A|-1}$), ②, …, |S|

$$\sum_{s \in S} P_{a_0}(1, s) \times R_{a_0}(1, s)$$

$$\sum_{s \in S} P_{a_1}(1, s) \times R_{a_1}(1, s)$$

$$V^T(S) = \max_{a \in A} \sum_{s' \in S} P_a(S, s') R_a(S, s')$$

$$\pi^T(S) = \operatorname{argmax}_{a \in A} \sum_{s' \in S} P_a(S, s') R_a(S, s')$$

Suppose now you're in second last round in state s

$\cancel{V^T(s)}$?

$$V^{T-1}(s, a) = \sum_{s' \in S} P_a(s, s') \left[V^T(s') + R_a(s, s') \right]$$

$$V^{T-1}(s) = \max_a V^{T-1}(s, a)$$

Inductively we can define

$$V^{T-K}(s) = \max_a \left[\sum_{s' \in S} P_a(s, s') \left\{ V^{T-K+1}(s') + R_a(s, s') \right\} \right]$$

Will this work if we have ∞ horizon?

0 1

a_0, a_1

$\pi(0) = a_0$ $P_{a_0}(0,1) = 0.6 = P_{a_1}(1,0)$

$\pi(1) = a_1$ $P_{a_0}(0,0) = 0.4 = P_{a_1}(1,1)$

For simplicity

$$R_{a_0}(0,1) = R_{a_0}(0,0).$$

$$R_{a_1}(1,0) = R_{a_1}(1,1) = 1$$

At some point

$V(0) = 2$ $V(1) = 3$

According to this policy, we update

$$V^{\pi}(0) = 0.4\,[1 + \overset{V(0)}{2}] + 0.6\,[1 + \overset{V(1)}{3}]$$

$$= 3.6$$

V_1 estimate:

$$V^{\pi}(1) = 0.4[1+\overset{V(1)}{3}] + 0.6[1+\overset{V(0)}{3.6}]$$

$$= \cancel{1.6}\ 4.36$$

As you keep play, assigning infinite rewards to states. Not easy to compare infinite rewards..

Say:

5, 5, 5, 5, 0, 5, 5, 5, 5, 5, 0, ...

4 4 4, 4, 4, 4 ...

Are both equal?

Sum is ∞ in both cases / How to compare

0, 0, 0, 0, 0, 5, 5, 5, 5, 0 ...

4, 4, 4, 4, 4, 4 ...

What about this?

Depends. Initially better vs initially worse.

We discount future by γ

Usually $0 < \gamma < 1$

How much you value future rewards today.

$$V^{\pi}(0) = 0.4[1+\gamma 2] + 0.6[1+\gamma 3]$$

$\cancel{V^{\pi}(1) = 0.4(1+}$

classmate
Date
Page

With that:

$$V(s) = \max_a \sum_{s' \in S} P_a(s, s') \times \left\{ R_a(s, s') + \underbrace{\gamma}_{\text{Discount factor}} V(s') \right\}$$

Two ways:

Value iteration:

$$V^{t+1}(s) = \quad V^t(s)$$

as $t \to \infty$

$$\| V^{t+1}(s) - V^t(s) \|_2 \to 0$$

As long as $\gamma < 1$, values will be fine & algorithm will converge

Policy iteration value:

Look at policy rather than value.

Practically, we may not always know what the rewards are & transition probabilities may not be known.

Using Monte Carlo Methods, if we know

the transition prob., not reward, do a large number of simulations & determine policy.

But may not know both. Maintain a Q-Learning table. ~~based on~~
SARSA

May sometimes be unstable, slight changes in rewards change policy.
Then:
Double Q methods where one table updates slower than the other
DQN
Double DQN.
DDPG popular neural network based algo.
Used to train AlphaZero.

Bellman backward induction?
<SARP> must be known for this & for value iteration. (for our case)

Next class Multi-arm bandit.
Short class 30-40min
Early attendance.

4-25 Sujit class 3

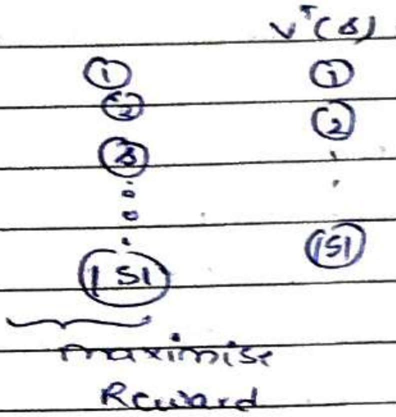
Recap:

Markov Decision Process (MDP)

S, A, R, P

Finite Horizon MDP Use MEU $v^T(S)$

$v^T(s)$



$v^{T-1}(s, a) = R$

$$\sum_{s'} \{R_a(s, s') + \underbrace{v^T(s')}_{\gamma}\} P_a(s, s')$$

Bellman Backward Induction.

Multiply by discount factor.

$$v^{t+1}(s) = \max_a \sum_{s'} P_a(s, s') \{ R_a(s, s') + v^t(s') \}$$

One special case of this becomes.

$|S| = 1$

$$\pi^*(s) = \arg\max_a R_a(s, s)$$

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Date
Page

$R_a = 1$ w.p. μ_a w.p. = with prob
$= 0$ otherwise

$a^* = \arg\max \mu_a$

what if you don't know μ_a

Multi-armed Bandits (MAB)

K actions
Greedy Approach
for $i = 1 \to m k$
use action i
Observe R_i
update $\hat{\mu}_i$
} Exploration

for $i = mk + 1 \to T$
use action $i^* = \arg\max \mu_i$
update $\hat{\mu}_i^*$
} Exploitation

Say I take actions in RR fashion.
$m = 1$
Issues: Suppose 2 actions
$\mu_1 = 0.9$
$\mu_2 = 0.8$

Probability of 0 reward = 0.1
with prob 0.8
$\hat{\mu}_1 = 0$ $\hat{\mu}_2 = 1$

If $m = T/K$

Half the round

$\left[\frac{0.1 \times T}{2}\right]$ loss on average

m should not be too large or small

In MAB, balance explo - ration / itation.

$m = 1$ called ϵ_t Greedy Algorithm

Exploitation

wp $(1-\epsilon)$, use greedy strategy

wp ϵ select an action randomly

(wp $1/k$)

How to set ϵ?

If very large - almost on average m to be order T.

Most time spent ~~to~~ exploration. Good learning

ϵ too small - no good learning - use second or third best arm.

Instead use

$$\epsilon_t = O\left(\frac{1}{t}\right)$$ Order $(1/t)$

As time ↑, ϵ ↓

ϵ can also be called learning rate.

Initially learn, then learn less & exploit

To compare algorithms:

Regret (Finite Horizon)

$$\sum_{t=1}^{T} \mu_{a^*} - \mu_{a^t}$$

Info theoritic lower bound.
No algo can do better than:
Regret has to be atleast

$\Omega(\log T)$ lower bound

Upper Confidance Bound (UCB)
Regret very close to $\Omega(\log T)$

$\log T (\sqrt{\log T})$?

Algorithm:
for $i = 1 \to k$
use (pull arm) action t
Update $\hat{\mu}_t$

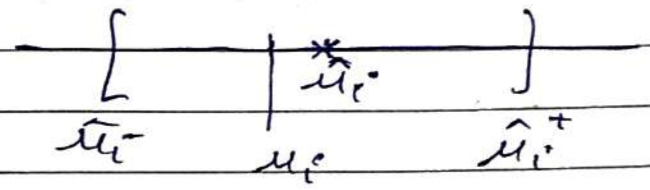
for $t = k+1 \to T$
pull $i^t = \text{argmax} \; \overbrace{\hat{\mu}_i^t \pm \sqrt{\frac{2 \log t}{N_{i,t}}}}^{UCB}$

$N_{i^t, t+1} = N_{i^t, t} + 1$

Can also be written as $\hat{\mu}_i^{\pm}$

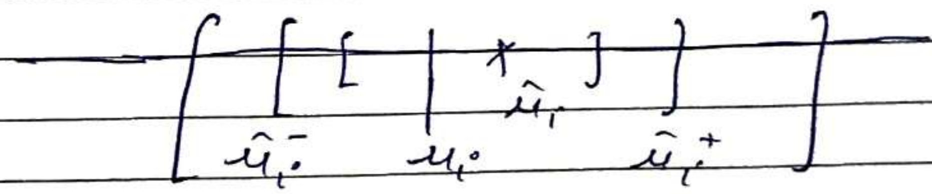
$N_{i,t}$:= # of times action/arm i is used till time t.

Intuitively



We can argue that supoptimal arm happened only log T times

μ_a^* $a^t \neq a^* | + O(\log T)$



$|\hat{\mu}_i^+ - \hat{\mu}_i|$ $|\mu_i^+ - \hat{\mu}_i|$?

Argue for each arm
So Regret $(k-1) \log T$

For optimal arm $N_{i,t}$ increases linearly(?)
$\log T$

E learning - special case of Q-learning
Q tables

$$Q(s_t, a_t) = R_{a_t}(s_t, s_{t+1}) + \max_a Q(s_{t+1}, a)$$

Update μ_i

$$\mu_{i,t+1} = \frac{\mu_{i,t} \times N_{i,t} + R_{i,t}}{N_{i,t} + 1}$$